

Tiling?

GPU's SRAM $\rightarrow$ very $\uparrow$ bandwidth
A very $\downarrow$ storage

So we use techniques to reduce size

B

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

$$C = A \times B$$

A

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

C

C_{11}	C_{12}		
C_{21}	C_{22}		

$$C_{1,1} = 1 \times 1 + 2 \times 5 + 3 \times 9 + 4 \times 13$$

$$C_{1,2} = 1 \times 2 + 2 \times 6 + 3 \times 10 + 4 \times 14$$

$$C_{2,1} = 5 \times 1 + 6 \times 5 + 7 \times 9 + 8 \times 13$$

$$C_{2,2} = 5 \times 2 + 6 \times 6 + 7 \times 10 + 8 \times 14$$

Takes 1 row from A, 1 col from B

$$4 \text{ mul} \rightarrow 4 \times 2 \rightarrow 8$$

each has 4 values $\Rightarrow$ 32 accesses

$$C_{11} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \end{bmatrix} + \begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} 9 \\ 13 \end{bmatrix}$$

$$C_{12} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 6 \end{bmatrix} + \begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} 10 \\ 14 \end{bmatrix}$$

$$\therefore \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 9 & 10 \\ 13 & 14 \end{bmatrix}$$

4 matrix with 4 values each

$\rightarrow$ 16 memory access.

for an $N \times N$ block done

memory access $\rightarrow 1/N$

Online Softmax:

a complex algo that came in 2018
writes softmax as:

$$d_i' = d_{i-1}' e^{m_{i-1} - m_i} + e^{x_i - m_i}$$

$$m_0 = -\infty$$

$$d_0 = 0$$

for $1 \leq i \leq N$ do:

$$x_i = q k_i^T$$

$$m_i = \max(m_{i-1}, x_i)$$

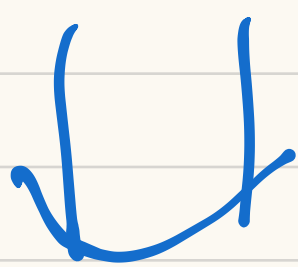
$$d_i' = d_{i-1}' e^{m_{i-1} - m_i} + e^{x_i - m_i}$$

$$o_0 = 0 \leftarrow$$

for $1 \leq i^R \leq N$ do

$$d_i = e^{x_i - m_N} \cdot d_i'$$

$$o_i = o_{i-1} + a_i v_i$$



FLASH

ATTENTION

Combine all into one single loop
→ reduce all matrix by fusing all computation together



all O_i for given q computed by
loading K_i & V_i for each token in
SRAM & save / write onto HMB

HARDWARE

FA2, FA1

ALGOS

TOUGHNESS :

- 1) implementation complexity
- 2) no access to weights (if needed for eviction policy)
- 3) Hardware specific code